CDAC-Mumbai
Fundamentals of AI and Mathematics

Date: 01-04-2026 **Total Marks:20**

Time duration: 1.00 hours. **Batch: Feb26**

Question Title: Image Compression and Feature Extraction using SVD and Gradient Optimization. [20 Marks]

Problem Statement

In many AI applications such as image processing and data compression, **matrix decomposition and optimization techniques** are used to reduce data size while preserving important information.

You are given a grayscale image represented as a matrix. Your task is to:

1. Perform **matrix decomposition using Singular Value Decomposition (SVD)**
2. Reconstruct a compressed version of the image
3. Analyze reconstruction error
4. Optimize reconstruction using **gradient-based minimization**

Input

A matrix A representing a grayscale image:

```
A = [
  [52, 55, 61, 59],
  [49, 50, 60, 58],
  [48, 52, 62, 60],
  [47, 49, 59, 57]
]
```

Tasks

Part A: Matrix Fundamentals (5 Marks)

1. Convert the input into a NumPy matrix
2. Compute:
   - Transpose of matrix
   - Determinant (if applicable)
   - Rank of matrix

Part B: Singular Value Decomposition (SVD) (5 Marks)

1. Perform SVD:

$$A = U\Sigma V^T$$

2. Extract singular values
3. Reconstruct matrix using top **k = 2** singular values

Part C: Error Analysis (5 Marks)

1. Compute reconstruction error using:
   - Mean Squared Error (MSE)
2. Compare original and compressed matrices

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Part D: Optimization using Gradient Descent (5 Marks)

Minimize reconstruction error:

$$Loss = || A - \hat{A} ||^2$$

Steps:

1. Initialize approximation matrix randomly
2. Use gradient descent to minimize error
3. Update matrix iteratively
4. Output optimized matrix

----------All the Best----------

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